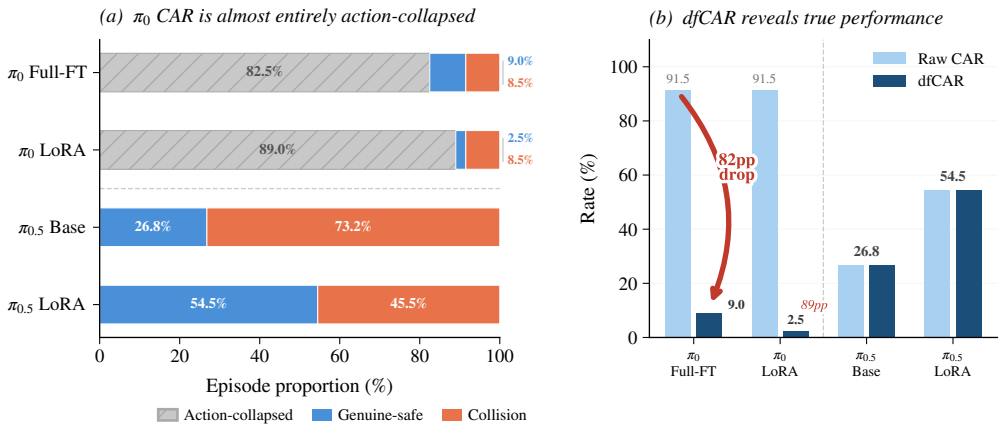
(a) π0 CAR is almost entirely action-collapsed
π0 Full-FT
π0 LoRA
π0.5 Base
π0.5 LoRA
82.5%
9.0%
8.5%
89.0%
2.5%
8.5%
26.8%
73.2%
54.5%
45.5%
0
20
40
60
80
100
Episode proportion (%)
Action-collapsed
Genuine-safe
Collision
(b) dfCAR reveals true performance
Rate (%)
Raw CAR
dfCAR
91.5
91.5
82pp drop
9.0
2.5
89pp
26.8
54.5
π0 Full-FT
π0 LoRA
π0.5 Base
π0.5 LoRA